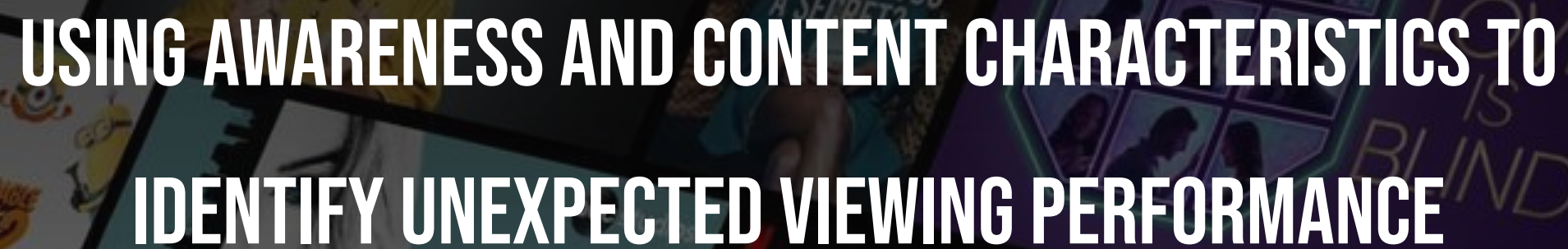


FINDING HIDDEN DEMAND IN NETFLIX'S TV CATALOG



USING AWARENESS AND CONTENT CHARACTERISTICS TO IDENTIFY UNEXPECTED VIEWING PERFORMANCE

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MEET OUR TEAM



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BUSINESS PROBLEM

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VIEWING HOURS DO NOT TELL THE WHOLE STORY.

How can Netflix distinguish normal performance from unusual audience conversion?

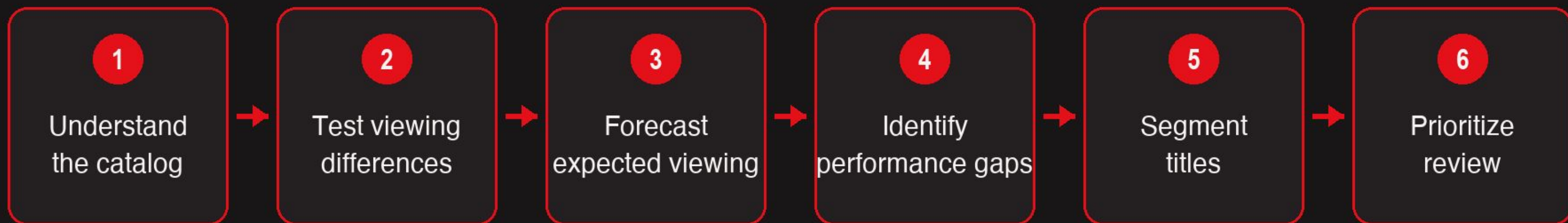
Netflix needs to know:

- Which titles outperform expectations?
- Which titles have unrealized demand?
- Where should limited review resources go?

LET'S START!

WORKFLOW AND METHODS

ANALYSIS WORKFLOW



DATA AND PREPARATION

1



2021 STREAMING
CATALOG

RATINGS • PLATFORM
AVAILABILITY • RELEASE YEAR

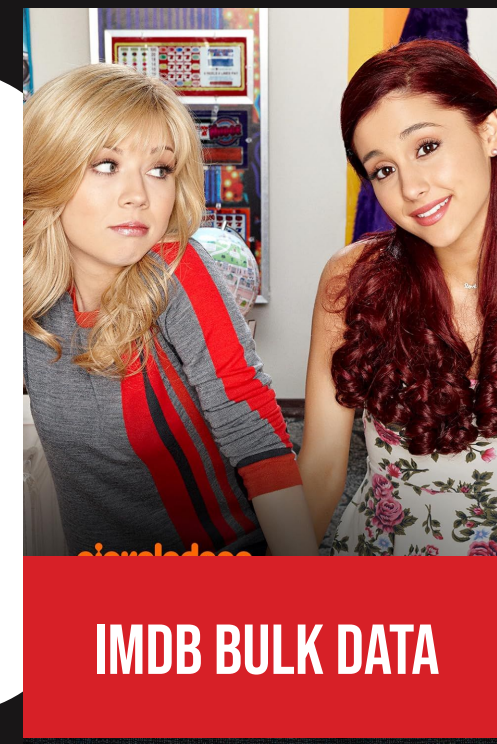
2



2021 IMDB RATING
COUNTS

ESTABLISHED AUDIENCE
AWARENESS • TV SHOWS ID

3



IMDB BULK DATA

ONLY FOR MATCHING TV SHOWS ID

4



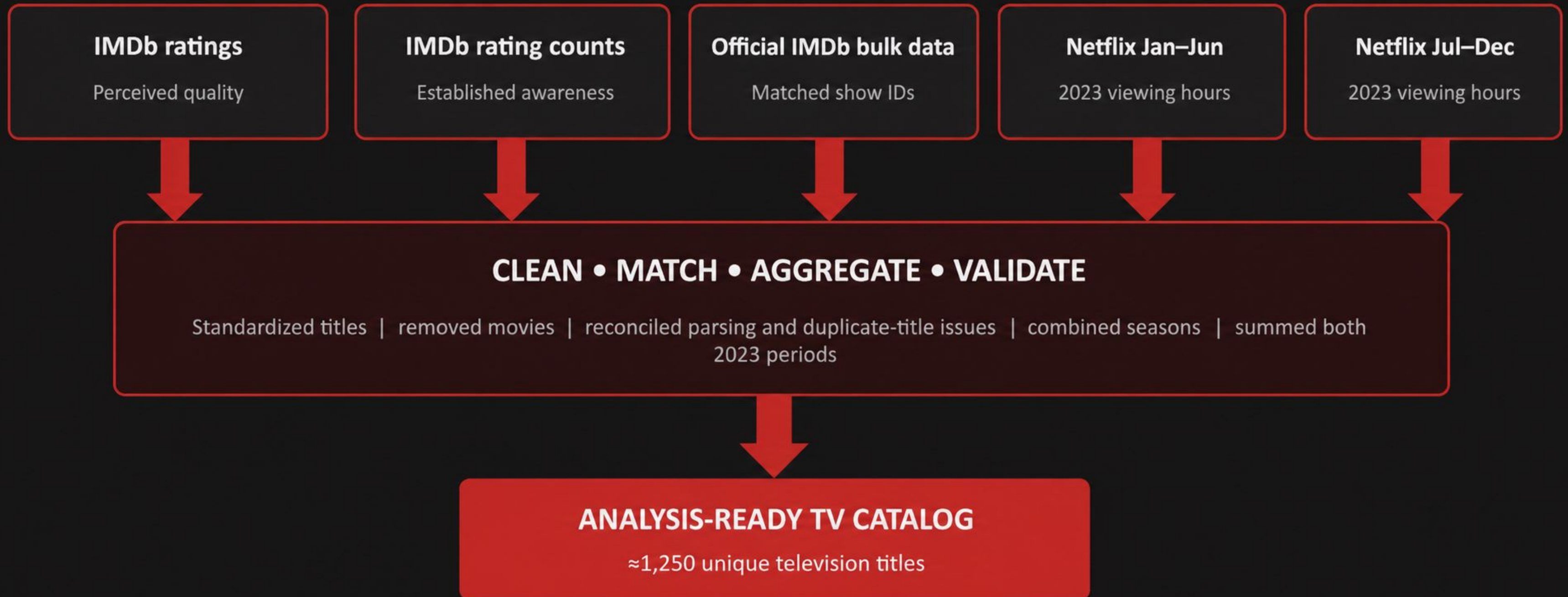
2023 NETFLIX
ENGAGEMENT REPORTS

ACTUAL HOURS VIEWED • GLOBAL
AVAILABILITY

1250+ MATCHED SERIES

Combined historical catalog characteristics with IMDb awareness data
and Netflix's actual 2023 engagement data.

DATA MERGING & CLEANING



VARIABLES

ANALYSIS ROLE	VARIABLES
Actual engagement	2023 Netflix viewing hours
Established awareness	IMDb rating count
Rating quality	IMDb rating • Rotten Tomatoes score
Content characteristics	Seasons • Title age • Series format • Audience category
Catalog context	Global availability • Historical platform status

KEY TERMS

Important terms to know

TERM	MEASURE	WHAT IT MEANS
Established awareness	IMDb rating count	How much public attention a title had accumulated
Expected viewing	Model benchmark	Viewing expected for a title with similar awareness and content
Out-of-fold prediction	Unseen-title estimate	Each title is predicted by a model that did not train on it
Performance gap	Actual minus expected	Positive = more viewing than expected; negative = less

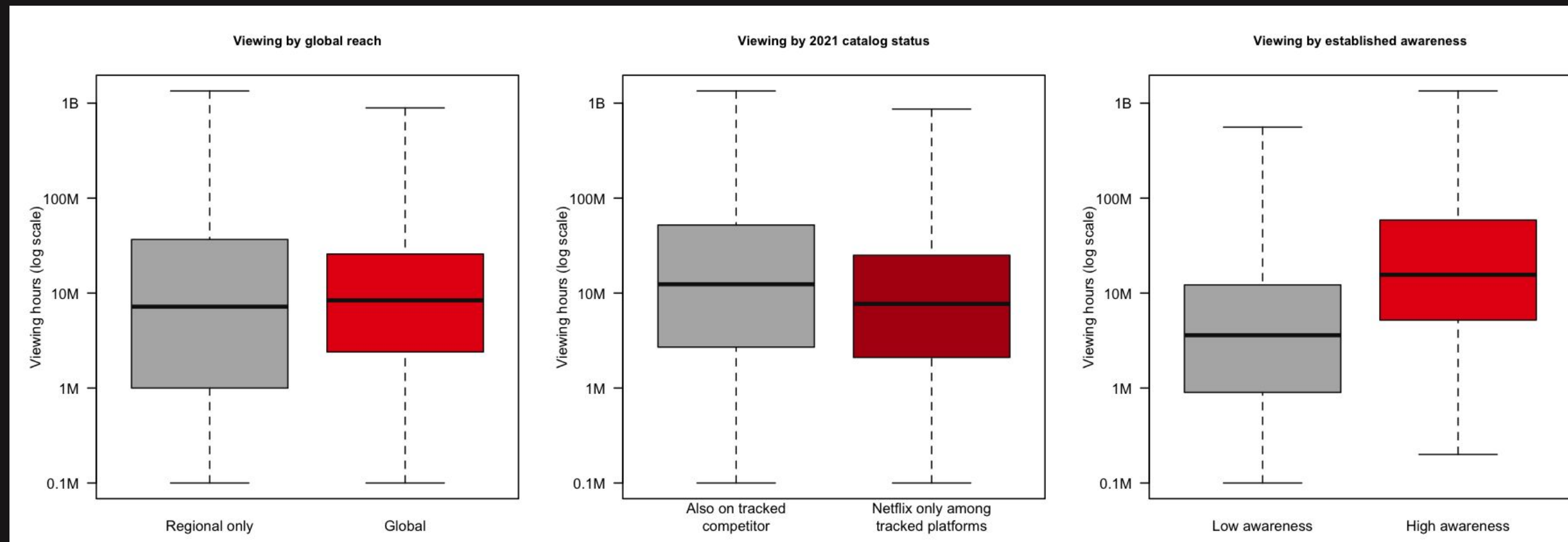
ANALYSIS

WELCH TWO SAMPLE T-TESTS

TEST BASIC VIEWING DIFFERENCES

Welch t-test results

COMPARISON	VIEWING RATIO	ADJUSTED P	CONCLUSION
Global vs regional	1.30×	0.077	No clear difference
Single-platform vs shared	0.65×	0.077	No clear difference
High vs low awareness	4.90×	<0.001	Clear difference



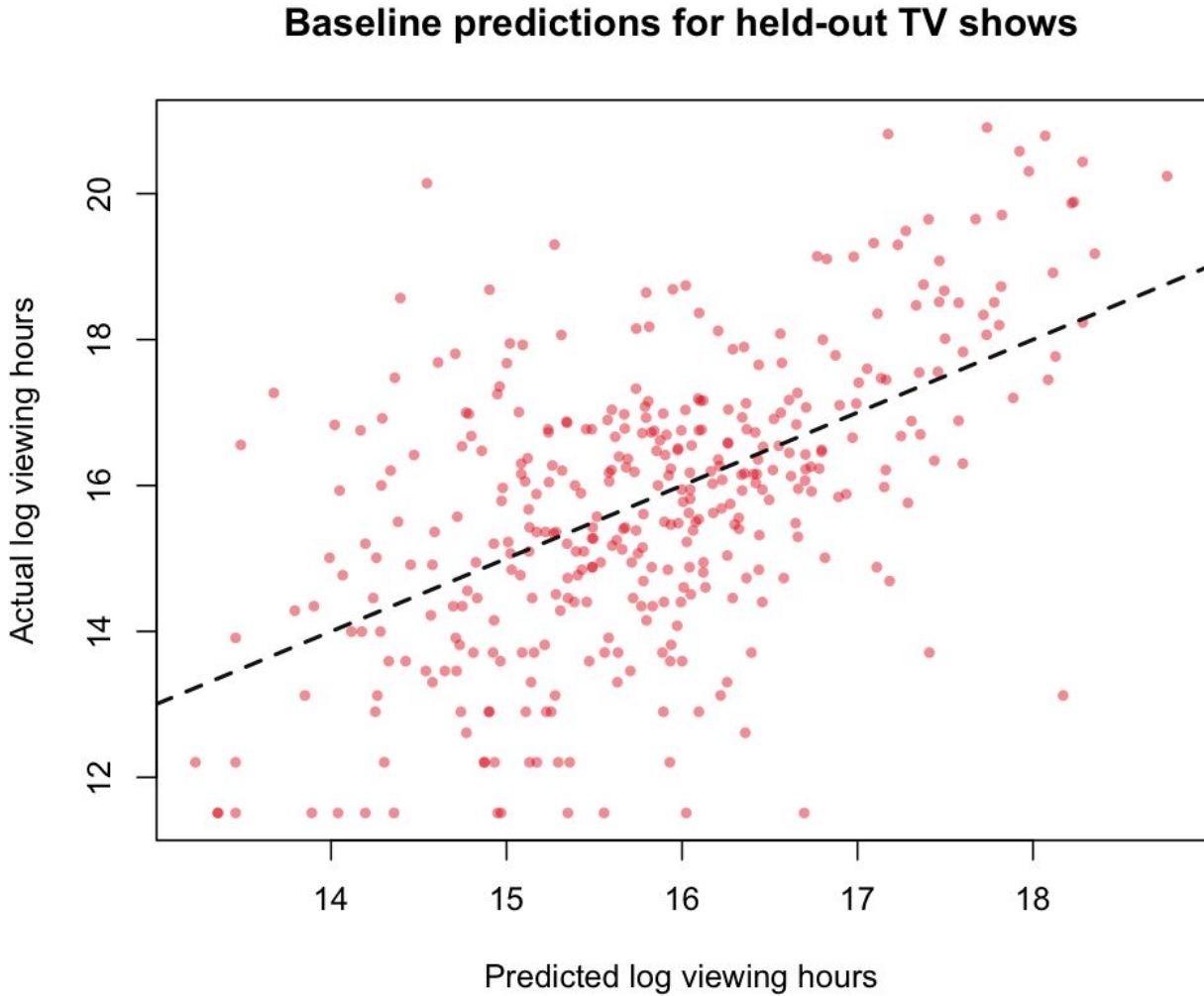
- High-awareness titles received approximately **4.9×** more viewing
- Availability and historical catalog position showed **no clear differences**
- Awareness provides the strongest starting point for predicting viewing

SIMPLE LINEAR REGRESSION

BUILD AN AWARENESS BASELINE

Awareness provides a useful—but incomplete—baseline

ITEM	RESULT	PLAIN-LANGUAGE MEANING
Model	Simple linear regression	—
Outcome	Log Netflix viewing hours	Continuous engagement benchmark
Predictor	Log IMDb rating count	Established awareness
Held-out titles	371	Not used to fit the model
Test R ²	0.308	Awareness explains part—not all—of viewing
Test correlation	0.56	Moderate agreement



IMDb awareness provides a useful baseline, but approximately 69% of test-title viewing variation remains unexplained.

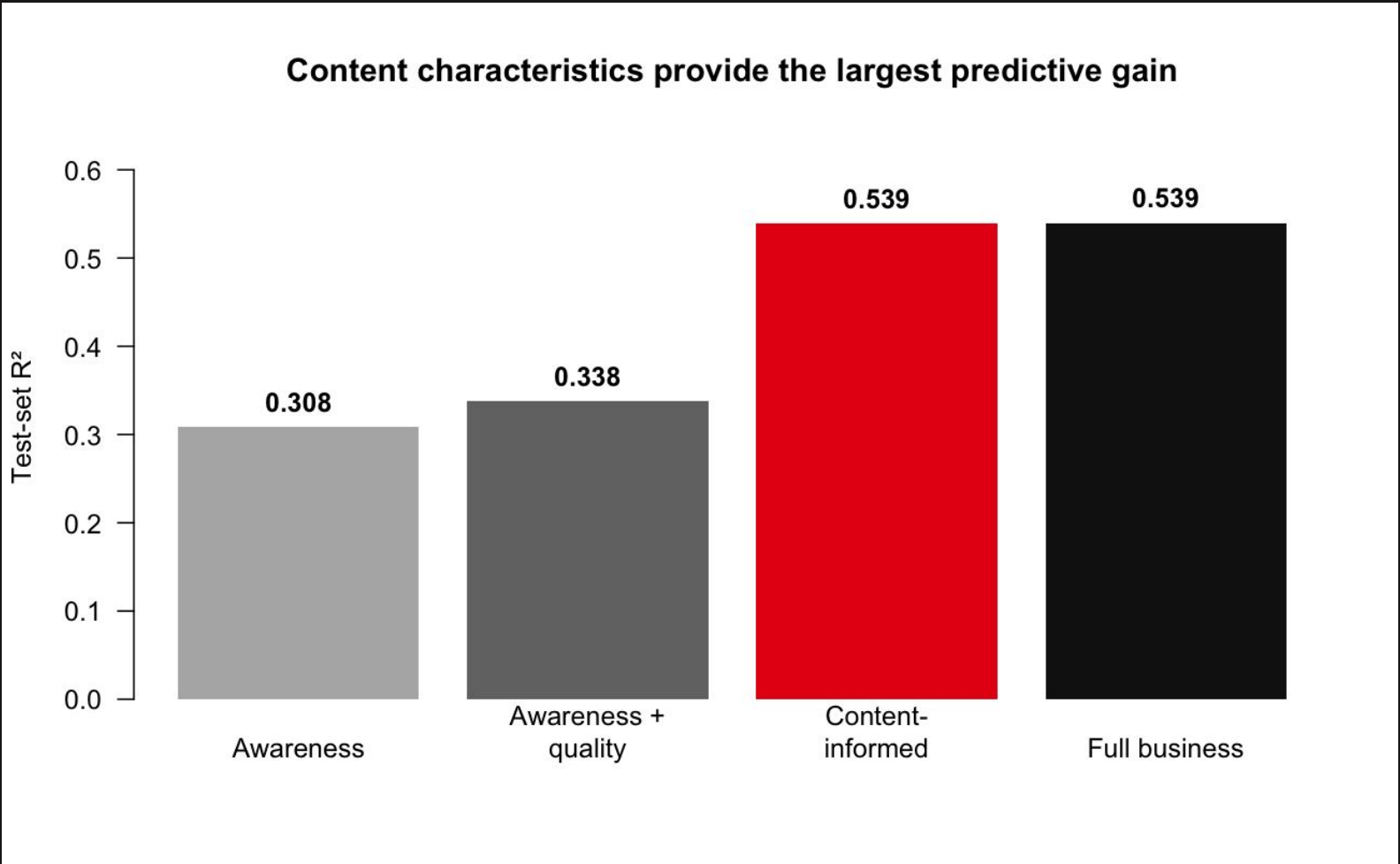
MULTIPLE LINEAR REGRESSION

IMPROVE THE FORECAST

Content characteristics create the strongest practical benchmark

MODEL	INFORMATION ADDED	TEST R²	WHAT CHANGED
Awareness baseline	IMDb rating count	0.308	Starting benchmark
Add rating quality	Awareness + IMDb/RT scores	0.338	+0.030 R²
Content-informed	+ seasons, age, format, audience	0.539	+0.201 R²
Full business model	+ availability and catalog status	0.539	No meaningful gain

Content Characteristics: seasons, title age, series format, audience category



Model progression:

1. Awareness
2. Awareness + rating quality
3. Awareness + content characteristics
4. Full model with catalog context

70/30 TRAINING-TESTING SPLIT

MODEL ACCURACY

Benchmark quality on 371 unseen titles

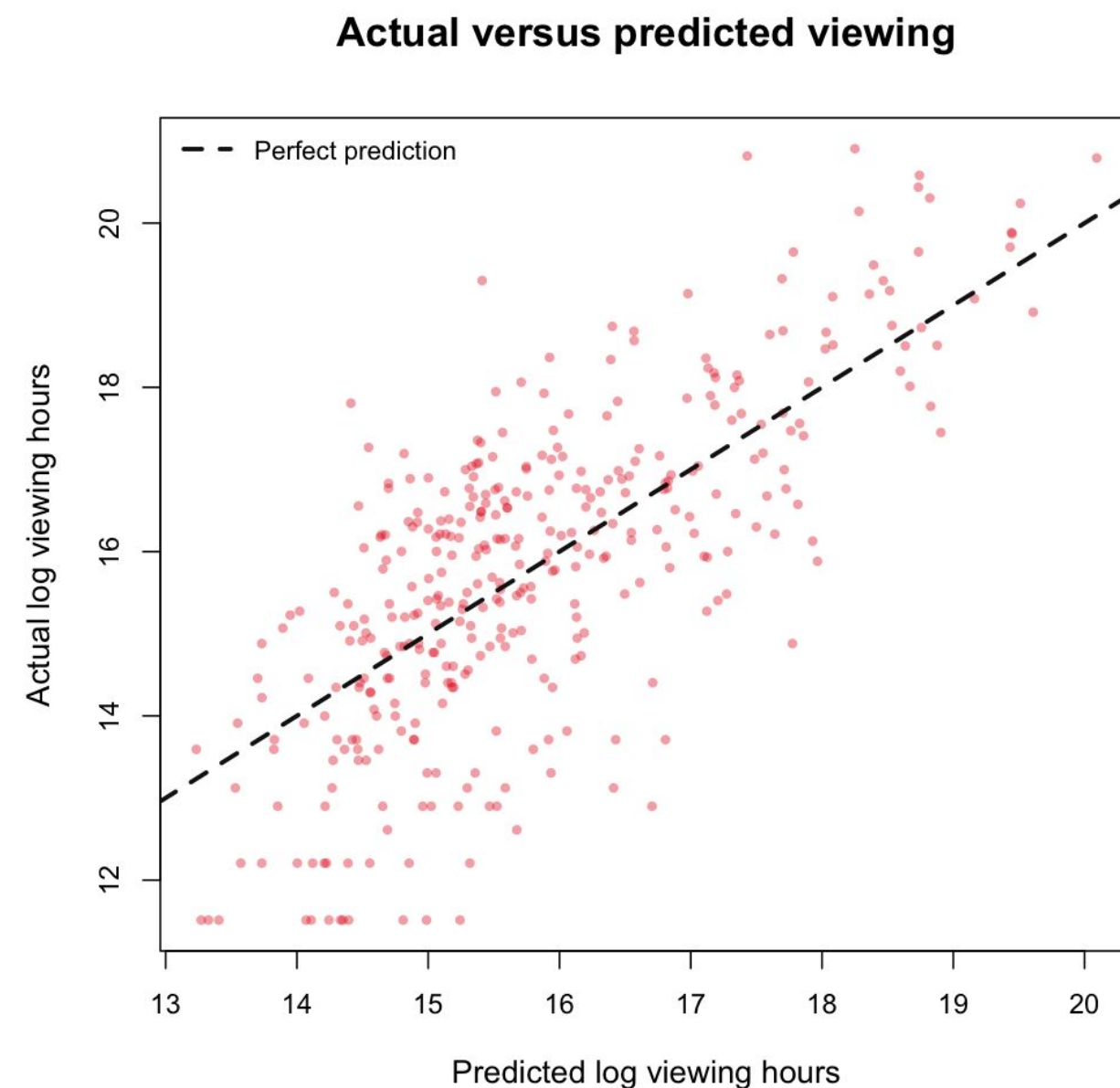
54% of viewing variation explained

0.74 actual-versus-predicted correlation

+75% R^2 improvement over awareness alone

Held-out test set: 371 titles

Use to prioritize review—not automate title decisions.



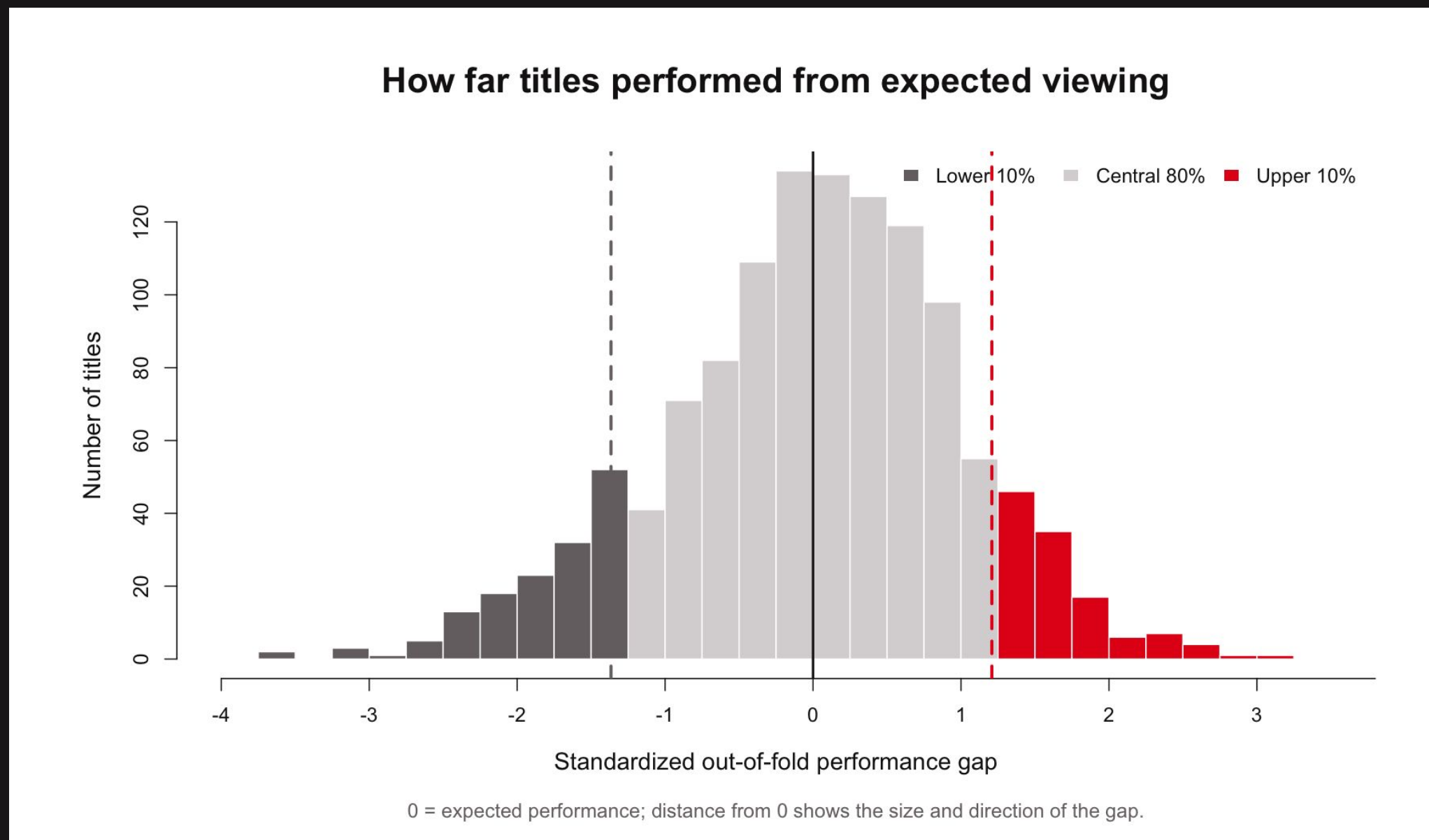
The remaining spread represents demand the available variables do not explain.

- Used 70% of the data to train the model and the other 30% to test it.
- The metrics shown use **371 unseen titles**.
- The dashed diagonal represents perfect predictions.
- Using 123 as the seed.
 - The dashed diagonal represents perfect predictions.

KEY FINDINGS

FINDINGS

MOST TITLES PERFORM NORMALLY



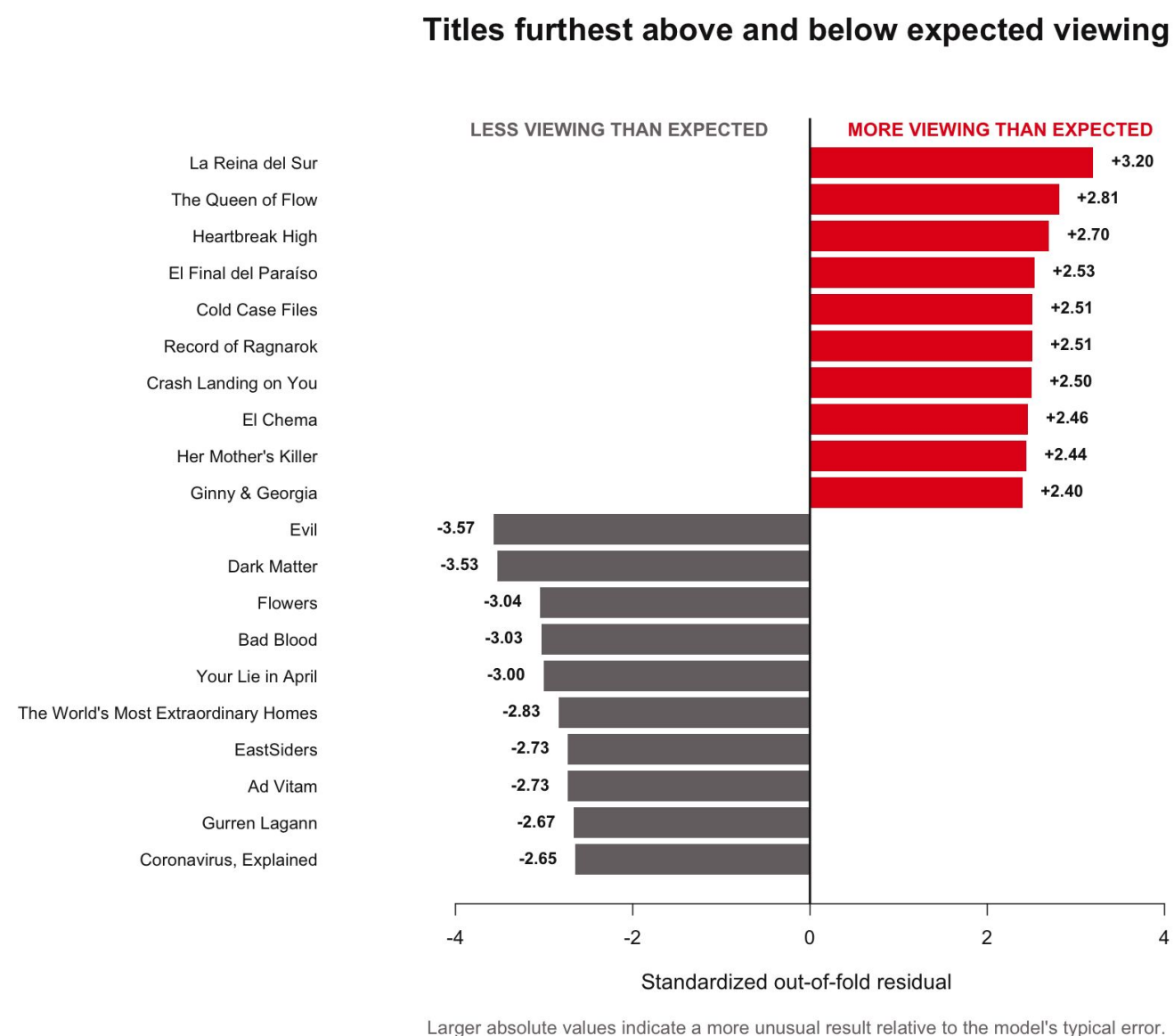
Performance Gap =

Actual Viewing - Expected Viewing

- Most titles cluster around expected viewing
- Distance from zero shows the size of the performance gap
- Both tails contain unusually large departures
- Tail titles become the priority review queue

FINDINGS

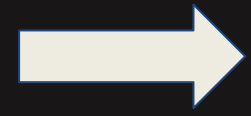
TITLES FURTHEST FROM EXPECTATIONS



- Performance gaps compare **actual viewing with expected viewing**
- *La Reina del Sur* showed the largest positive gap
- *Evil* and *Dark Matter* showed the largest negative gaps
- Large gaps identify titles that deserve deeper investigation
- Results are signals, **not explanations of performance**

BUSINESS IMPLICATIONS

VIEWING DATA



FOCUSED ACTION

1



AMPLIFY
titles with
unexpected
emerging demand

2



PROTECT
titles with
established engagement

3



INVESTIGATE
titles with
unrealized
audience potential

USE THE MODEL AS A REVIEW

	 Licensed Shows	 Netflix Originals
 ABOVE EXPECTED	 Explore licensing & promotion Review broader distribution	 Explore promotion or renewal Consider continued investment
 NEAR EXPECTED	 Monitor value & cost Compare performance with price	 Maintain & monitor Track continued engagement
 BELOW EXPECTED	 Review licensing value Recheck audience fit	 Investigate before investing Review positioning and response

A screening tool—not an automatic decision. Add cost, retention, promotion, and strategic fit.

RESOURCE PAGE

TV Shows on Netflix, Prime Video, Hulu, and Disney+

<https://www.kaggle.com/datasets/ruchi798/tv-shows-on-netflix-prime-video-hulu-and-disney/data>

Netflix engagement report:

- <https://about.netflix.com/en/news/what-we-watched-the-second-half-of-2023>
- <https://about.netflix.com/en/news/what-we-watched-a-netflix-engagement-report>

IMDb 2021 rating count data (2021 Official IMDb bulk data snapshot):

<https://www.kaggle.com/datasets/valchovalex/imdb-title-ratings-2021>

Official IMDb bulk data:

<https://datasets.imdbws.com/>